Muzaddidul Islam Chowdhury



I want to research how data and AI can carry clean-energy materials from the lab into reliable, large-scale production.

Open to funded PhD positions for 2027 and 2028 intakes. plabon.mic@gmail.com | linkedin.com/in/muzaddidul | scholar.google.com/citations?user=mdswSJ8AAAAJ

2018–19
Lab
$BiFeO_3$ nanoparticles for solar cells

2018–21
Devices
Carbon-nanotube FET modelling, 2 IEEE papers

2021–22
Storage
MXene battery-electrode review

2021–23
Factory
Process Analyst, production engineering at BAT

2023–25
Supply network
Scheduler for a USD 67M+ material portfolio at BAT

2025–now
AI for manufacturing
Project Manager at Aivelon

I have synthesised and characterised bismuth-ferrite photovoltaic materials and co-authored work on carbon-nanotube transistors and MXene battery electrodes. Then I spent four years inside a multinational manufacturer, first in production engineering and then running a USD 67M+ material supply network at British American Tobacco. Now I lead AI software delivery for manufacturing clients at Aivelon. I fit labs working on battery or solar manufacturing, process analytics, or supply chains for energy materials.

5 publications in IEEE, IOP and JNEP venues on photovoltaic oxides, nanotube transistors and battery electrodes.

4+ years in a multinational factory, from spare parts for the production line to scheduling imported materials.

Builds working tools: MRP automation, Power BI safety-stock models, a currency-risk app used through a forex crisis.

What I want to research

1 **Learning the process window for materials scale-up**

In my $BiFeO_3$ work, the annealing temperature alone decided whether secondary phases appeared and where the bandgap landed. At production scale, that window becomes a yield and cost problem. I want to build models that link synthesis and process parameters to material quality, so a plant can hold the window under real variation.

2 **Quality and yield analytics in battery and solar cell manufacturing**

Inline sensing, defect detection and digital twins for electrode and cell production: the factory-side version of the electrode materials I reviewed in the MXene paper, using the plant data habits I built at BAT.

3 **Resilient supply chains for energy materials**

At BAT I watched a small demand shift in one shared raw material cascade into a bullwhip that nearly stopped the line, and I built a currency-risk tool to keep imports flowing in 2023. Battery and PV materials face the same volatility with higher stakes. I want to model sourcing, inventory and risk for them.

Publications

MXene based heterostructures for electrode materials of batteries: a review
M. A. H. Shanto, **M. I. Chowdhury**, A. B. Antu, N. R. Niloy, N. Alam, M. A. Ullah, M. S. Anowar
IOP Conference Series: Materials Science and Engineering, vol. 1225, 012018 (IC-MAME 2021)
Open access
Journal-indexed proceedings, 2022 DOI

A study of OFF-state current and impact of dielectric constant on I_{ON}/I_{OFF} ratio of carbon nanotube field effect transistor
M. A. H. Shanto, **M. I. Chowdhury**, N. R. Niloy, N. Alam, M. J. Nayem, S.-A. Imam
2021 6th International Conference for Convergence in Technology (I2CT), Pune, India. IEEE
IEEE conference, 2021 **DOI** **PDF**

Multiferroic bismuth ferrite nanocomposites as a potential photovoltaic material
N. R. Niloy, **M. I. Chowdhury**, M. A. H. Shanto, J. Islam, M. M. Rhaman
IOP Conference Series: Materials Science and Engineering, vol. 1091, 012049 (ICIRMCT 2021)
Journal-indexed proceedings, 2021 **DOI** **PDF**

Structural and optical characterization of multiferroic $BiFeO_3$ nanoparticles synthesized at different annealing temperatures
N. R. Niloy, **M. I. Chowdhury**, S. Anowar, J. Islam, M. M. Rhaman
Journal of Nano- and Electronic Physics, vol. 12, no. 5, 05015 (2020)
Journal article, 2020 **DOI** **PDF**

Effect of channel length and dielectric constant on carbon nanotube FET to evaluate the device performance
M. A. H. Shanto, N. R. Niloy, R. Islam, **M. I. Chowdhury**, M. M. Abrar, S.-A. Imam
2020 4th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech), Kolkata, India, pp. 1–4. IEEE
IEEE conference, 2020 **DOI** **PDF**

Research experience

Properties of multiferroic bismuth ferrite nanoparticles and their prospect in photovoltaic applications Jul 2018 – Aug 2019
Undergraduate thesis, AUST. Supervisor: Dr. Md. Meganur Rhaman, Professor, EEE

Synthesised $BiFeO_3$ nanoparticles by the sol-gel method and annealed them at 500, 600 and 700 °C. Characterised structure with XRD, FESEM, TEM and EDS, and optical properties with UV-Vis-NIR. The 500 °C samples showed fewer secondary phases and a favourable bandgap for absorber layers. Led to the JNEP 2020 and IOP 2021 papers.

Improving the performance of single-wall carbon nanotube FETs Mar 2018 – Jun 2019
Undergraduate research, AUST. Supervisor: Safayat-Al Imam, Assistant Professor, EEE

Modelled non-ballistic effects (scattering, strain and tunnelling) in CNT-FETs. Showed how high-κ gate dielectrics strengthen electrostatic control, shorten the usable channel and suppress off-state current. Led to the IEMENTech 2020 and I2CT 2021 papers.

Industry experience

Project Manager Nov 2025 – present
Aivelon, Dhaka

Software company building SaaS and generative-AI-as-a-service products, currently focused on manufacturing clients.

- Lead the delivery team and client relationships, turning manufacturing clients' operational problems into software requirements and delivery plans.
- Supervise an engineering intern and work through the backend architecture (Java, Spring Boot) with him.

Wrapping Material Scheduler, Supply Network Operations Mar 2023 – Oct 2025
British American Tobacco Bangladesh, Dhaka

Planned packaging materials for a plant producing about 60 billion sticks a year.

- Built and ran inventory models for a USD 67M+ material portfolio, reaching 85% forecast accuracy from demand and consumption data.
- Coordinated 10+ foreign suppliers across Asia and Europe, from import documents and letters of credit to port clearance.
- Built *LC Analyzer* in Power Apps to track foreign-currency exposure during the 2023 forex crisis; every letter of credit for material imports was executed.

Process Analyst, Production Engineering Jun 2021 – Feb 2023
British American Tobacco Bangladesh, Dhaka

Spare parts and maintenance readiness for the production line.

- Reduced overseas OEM spare-parts lead time from 88 to 71 days, improving maintenance readiness by 18%.
- Automated material requirement planning in Excel VBA, reaching 100% procurement accuracy.
- Built a Power BI model over 8,000+ spare-part types to set safety stock from consumption patterns at a 95% service level.

Maintenance Engineer Oct 2019 – Jul 2021
Apsara Trading Ltd., Dhaka

Tools I built in industry

- **LC Analyzer.** Power Apps tool that scored letter-of-credit and currency risk for every import order during Bangladesh's 2023 forex crisis.
- **Project Amazon.** Power Apps digital indent system that replaced paper approvals and cut approval cycles by 85%.
- **Project Hulk.** Excel decision-support tool using lead-time analytics to set min-max levels and raise stock-out alerts automatically.
- **Automated MRP.** Excel VBA material requirement planning for spare parts, removing manual ordering errors.

Education

B.Sc. in Electrical and Electronic Engineering 2015 – 2021
Ahsanullah University of Science and Technology (AUST), Dhaka

Major in electronics. Thesis on bismuth ferrite nanoparticles for photovoltaics.

Skills

Materials and devices

Sol-gel synthesis; XRD, FESEM, TEM, EDS, UV-Vis-NIR; FullProf, VESTA, OriginPro; COMSOL, Cadence Virtuoso, PSpice

Manufacturing and supply chain

SAP ERP, Kinaxis RapidResponse, MRP, EOQ, safety stock, Lean (VSM, loss deployment)

Data and software

Python, MATLAB, C++, Power BI, Power Apps, Excel VBA; working knowledge of Java and Spring Boot